

JOHN KIM

SENIOR SOFTWARE ENGINEER

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PROFESSIONAL OVERVIEW

Senior Full Stack Engineer (AI/LLM | Node.js | Cloud) with 9 years of experience designing and scaling AI-driven, Node.js-based applications across fintech, SaaS, and data intelligence sectors. Specialized in NestJS, Express.js, React, Next.js, and React Native, with advanced proficiency in MCP-integrated microservices, TensorFlow/LLM model deployment, and multi-cloud DevOps (AWS & GCP). Delivers production-grade, intelligent systems combining machine learning, automation, and full-stack architecture.

WORK EXPERIENCE

Senior Software Engineer

Kalshi | New York, NY

AUG 2023 - SEP 2025

- Led architecture and development of multi-region NestJS microservices handling tens of thousands of trades per second with sub-100ms latency and 99.99% uptime across AWS ECS and GCP Cloud Run.
- Designed data persistence layer using MongoDB as the core NoSQL database for trade storage, real-time event logging, and market analytics, optimizing read/write performance under high concurrency.
- Built AI-enhanced analytics microservices with LangChain and OpenAI GPT-4, enabling traders to query live market data using natural language and receive predictive insights.
- Integrated TensorFlow-based models for trade prediction and risk scoring, with retraining pipelines written in Python and MCP-governed model tracking to ensure compliance and bias transparency.
- Developed event-driven infrastructure using Kafka, Redis Streams, and Pub/Sub to synchronize microservices and enable multi-zone fault tolerance.
- Engineered Next.js + React dashboards rendering real-time insights and AI-generated summaries through Vertex AI APIs.
- Built React Native mobile apps using shared TypeScript modules with web components, reducing development redundancy by 35%.
- Deployed GraphQL federation over Node.js services to unify trading, AI, and analytics data, cutting query latency by 45%.
- Constructed AI middleware in Node.js that dynamically routed inference requests between OpenAI, Hugging Face, and Vertex AI, balancing accuracy and inference cost.
- Designed prompt-chaining and RAG (retrieval-augmented generation) pipelines using LangChain and Pinecone, integrating contextual embeddings directly with MongoDB documents.
- Created serverless AI pipelines (AWS Lambda & GCP Cloud Functions) that auto-summarized trades and compliance logs, reducing manual review time by 5,000 hours annually.
- Built Terraform-based CI/CD pipelines orchestrating Dockerized deployments and automated blue-green releases with full observability in OpenTelemetry and Sentry.
- Mentored a cross-functional team of 10 engineers on cloud scaling, Node.js optimization, and responsible AI engineering standards across multi-cloud environments.

EDUCATION

Bachelor's degree

Penn State University

2013 - 2017

SKILLS

Programming Languages & Frameworks

JavaScript (ES6+), TypeScript, Node.js, Python, SQL, Express.js, NestJS, Fastify, React, Next.js, React Native, Vue.js, Nuxt.js, HTML5, CSS3, Tailwind CSS, Material UI

Backend & API Development

Microservices Architecture, RESTful APIs, GraphQL (Apollo, Federation), gRPC, Event-Driven Systems, WebSockets, CQRS, Serverless (AWS Lambda, GCP Functions), Domain-Driven Design (DDD)

Frontend Engineering

React, Next.js (SSR/SSG), Vue.js, Nuxt.js, React Native (iOS/Android), Redux, React Query, Styled Components, Progressive Web Apps (PWA), Accessibility (WCAG), Performance Optimization

Databases & Data Management

MongoDB (Core NoSQL), PostgreSQL, MySQL, Redis, DynamoDB, Firestore, Elasticsearch, Prisma, Mongoose ORM, Caching Strategies, Indexing & Query Optimization, Data Modeling

AI / ML / LLM Engineering

TensorFlow, PyTorch, Scikit-learn, OpenAI GPT-4 API, LangChain, Vertex AI, Hugging Face Transformers, Pinecone, Weaviate, NLP, RAG (Retrieval-Augmented Generation), Model Fine-tuning, Prompt Engineering, Model Deployment, AI Observability, MCP Governance

Senior Software Associate

Capital One | Philadelphia, PA

JUN 2021 – JUL 2023

- Re-engineered legacy banking modules into Node.js (Express.js) microservices serving tens of millions of API calls daily with 99.9 % availability.
- Built AI-assisted credit-risk prediction engine using TensorFlow and Scikit-learn, deployed through Node.js wrappers to enable real-time model inference within lending APIs.
- Integrated LLM summarization services (OpenAI + Vertex AI) for automated regulatory report drafting and email triage within internal workflows.
- Designed MCP-compliant microservices to monitor and log AI model outputs for audit traceability and ethics oversight.
- Developed React dashboards for risk and fraud teams, visualizing TensorFlow predictions and real-time alerts from Kafka event streams.
- Created serverless batch processing via AWS Lambda and GCP Pub/Sub for large-scale model re-training and data ingestion.
- Established CI/CD pipelines (Jenkins, Terraform, AWS CodePipeline) enabling automated testing, model promotion, and production deployment.
- Partnered with InfoSec to enforce JWT/OAuth2 authentication and data encryption within AI microservices under PCI-DSS guidelines.
- Co-authored internal best-practice documentation for AI service integration and MCP data handling standards.

Software Engineer

JPMorgan Chase & Co | Philadelphia, PA

SEP 2018 – May 2021

- Developed Node.js REST APIs and Python data pipelines for enterprise risk analytics used by global investment teams.
- Created Vue.js dashboards with real-time visualizations of AI-driven risk scores and performance indicators.
- Integrated Python-based anomaly detection models with Express APIs to provide early warning signals for compliance departments.
- Deployed to AWS ECS/EC2 and automated build pipelines through Jenkins and CloudFormation.
- Optimized Redis and PostgreSQL queries, reducing average response times by 30 %.

Junior Software Developer

Zip Code Wilmington | Philadelphia, PA

May 2018 – AUG 2018

- Developed full-stack applications using Node.js, Express, and React, contributing to student management and performance tracking systems.
- Implemented responsive front-end interfaces with React Hooks and reusable components, ensuring consistent UI behavior across devices.
- Built and maintained RESTful APIs integrating PostgreSQL via Sequelize ORM for data persistence and analytics.
- Designed internal admin dashboards with role-based authentication and automated report generation.
- Managed CI workflows through Git, conducted peer reviews, and maintained Agile sprint documentation for delivery consistency.

Cloud Platforms & DevOps

AWS (ECS, Lambda, API Gateway, CloudWatch, S3, SNS/SQS), GCP (Cloud Run, Vertex AI, BigQuery, Pub/Sub, Cloud Storage, Cloud Build), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Nginx, CI/CD, Helm, Blue-Green & Canary Deployments, Cloud Security

System Design & Infrastructure

Distributed Systems, High Availability (HA), Horizontal Scaling, Load Balancing, Caching Layers, Rate Limiting, Fault Tolerance, Event Sourcing, Observability (OpenTelemetry, Grafana, Prometheus, Datadog)

Testing & Quality Assurance

Jest, Cypress, Playwright, Mocha, Chai, Supertest, Postman, TDD/BDD, Unit & Integration Testing, Regression Testing, Automation Pipelines

Security & Compliance

OAuth2, JWT, TLS/SSL, SAML, RBAC/ABAC, Data Encryption, PCI-DSS, SOC2, OWASP, Secure API Development, MCP-Compliant Data Governance

Tooling & Collaboration:

Git, GitHub, GitLab, Bitbucket, Jira, Confluence, Agile/Scrum, REST/GraphQL API Documentation (Swagger/OpenAPI), Postman, Slack Integrations

Professional & Leadership:

Team Mentorship, Technical Strategy, Cross-Functional Collaboration, Stakeholder Communication, Code Review Leadership, Architectural Decision-Making, Agile Coaching, Product Alignment, Continuous Improvement